

Project Documentation

Project: E-Commerce Sales Analysis

1. Problem Statement

An e-commerce company wants to analyze its sales performance, customer behavior, and product performance. The available data contains inconsistencies such as missing values, duplicates, inconsistent text formatting, and invalid records. The goal is to clean the data and create an interactive Power BI dashboard that provides business insights.

2. Business Objectives

- Improve data quality.
- Analyze sales performance.
- Identify top-selling and low-performing products.
- Understand customer purchasing behavior.
- Identify top customers and regions.
- Build an interactive Power BI dashboard.

3. Business Questions

Customer Analysis

- Who are the top customers?
- Which customer segment contributes the most revenue?
- Which age group contributes the most sales?
- Which gender contributes more revenue?
- Which customer segment is the most profitable?
- Which region contributes the highest revenue?

Order analysis

- How many total orders were placed?
- How many unique customers placed orders?
- How many unique products were ordered?
- What is the monthly trend of orders?
- Which month has the highest number of orders?
- Which customers placed the highest number of orders?
- Which products were ordered most frequently?
- Which customers have repeat orders?
- What is the average delivery time
- How many orders cancelled?
- How many orders returned?
- How many orders are delivered?
- Which month has the highest number of deliveries?
- Which month has the lowest number of deliveries?

Product Analysis

- Which products generate the highest sales?
- Which products have the highest profit?
- Which categories are most profitable?
- Which shipping mode is used most?
- Which payment method is most preferred?
- What are the total sales?
- Which category has the highest average profit margin?
- Which month has the highest sales?
- What are the monthly sales trends?
- Which regions have the highest profit?
- How often are sales targets achieved?

4. Scope

Use CSV files, Excel for profiling, Power Query for cleaning, Power BI for dashboarding.

5. Deliverables

Clean dataset, dashboard, insights, recommendations, project documentation.

Business Requirement Document (Template)

Field	Details
Stakeholders	Sales Team, Management
KPIs	Sales, Profit, Orders
Assumptions	Data is available and accurate after cleaning.
Risks	Missing or incorrect data.
Success Criteria	Dashboard is completed and provides useful insights.
Timeline	2 Weeks

Project Charter

Project Sponsor: Sales Director

Project Manager: Data Analytics Team

Duration: 2 Weeks

Tools: Excel, Power Query, Power BI

Data Collection & Understanding

Data Inventory

File	Purpose	Records	Columns
Customer.csv	Customer	1030	8
Orders.csv	Order	1030	9
Product.csv	Product	1030	3

Data Dictionary Template

Customer.csv

Column	Description	Data Type
Customer_ID	Unique identifier for each customer	Text
Customer_Name	Full name of the customer	Text
Customer_Segment	Category of the customer (e.g., Consumer, Corporate, Home Office)	Text
Gender	Gender of the customer	Text
Age_Group	Age group of the customer	Text
Region	Geographic region of the customer	Text
State	where the customer is located	Text
City	City where the customer is located	Text

Order.csv

Column	Description	Data Type
Order_ID	Unique identifier for each order	Text
Customer_ID	Unique identifier for the customer who placed the order	Text

Product_ID	Unique identifier for the product ordered	Text
OrderDate	Date when the order was placed	Date
Ship_Date	Date when the order was shipped	Date
Delivery_Status	Current delivery status of the order	Text

Column	Description	Data Type
Product_Name	Name of the product	Text
Unit_Price	Selling price of one unit of the product	Decimal Number
Cost	Cost price of the product	Decimal Number
Profit	Profit earned on the product	Decimal Number
Payment_Mode	Payment method used by the customer	Text
Shipping_Mode	Shipping method used for delivery	Text
Category	Main category of the product	Text
Sub_Category	Sub-category of the product	Text
Sales	Total sales amount for the transaction	Decimal Number

Excel Data Quality Checklist

- Check missing values
- Find duplicates
- Validate dates
- Check negative sales
- Verify data types
- Identify blanks

Data Understanding & Data Quality Assessment (Excel)

Objective

Profile the CSV data in Excel before any cleaning in Power Query.

Tasks

- Open all CSV files in Excel
- Check row and column counts
- Identify missing values
- Find duplicate records
- Validate date formats
- Check data types
- Identify outliers and negative values
- Document observations

Data Quality Report Template

Customer.csv

Issue	Column	Count	Action Required
Missing Values	Customer_Name	16	Fill missing names from source
Missing Values	City	14	Update missing city values from source data
Missing Values	Customer_ID	0	No action required
Missing Values	Customer_Segment	0	No action required
Missing Values	Gender	0	No action required
Missing Values	Age_Group	0	No action required
Missing Values	Region	0	No action required
Missing Values	State	0	No action required

- Trim extra spaces from all text columns.
- Convert **Customer_Name** to **Proper Case**.
- Check for **duplicate rows**

Order.csv

No Missing values

Product.csv

Issue	Column	Count	Action Required
Missing Values	Product_Name	0	No action required
Missing Values	Unit_Price	25	Fill missing prices from source data
Missing Values	Cost	0	No action required
Missing Values	Profit	0	No action required
Missing Values	Payment_Mode	11	Fill missing payment mode
Missing Values	Shipping_Mode	0	No action required
Missing Values	Category	17	Update missing category values from source data
Missing Values	Sub_Category	0	No action required
Missing Values	Sales	0	No action required

- Trim extra spaces from all text columns.
- Standardize text values (capitalize).

Data Exploration

Import data into Power BI

- Load – to import the data directly.
- Transform Data – to clean and prepare the data in Power Query before loading.
- multiple tables (Orders, Customers, Products), repeat the process for each table.

After loading, go to **Model View** and create relationships:

- **Orders[Customer ID] → Customers[Customer ID]**
 - **Orders[Product ID] → Products[Product ID]**
-
- Save the Power BI file (.pbix) and begin creating reports and dashboards.

Output

- Data imported successfully into Power BI.
- Tables available in the Fields pane.
- Relationships created in the Model view.
- Dataset ready for data cleaning, DAX measures, and visualization.

Recommended KPI Cards

Display these measures in **Card** visuals on dashboard:

- | | |
|--------------------|--|
| • Total Sales | Total Sales = SUM(Orders[Amount]) |
| • Total Orders | Total Orders = COUNT(Orders[Order ID]) |
| • Total Profit | Total Profit = SUM(Orders[Profit]) |
| • Unique Customers | Total Customers = DISTINCTCOUNT(Orders[Customer ID]) |
| • Maximum Sales | Maximum Sales = MAX(Orders[Amount]) |
| • Minimum Sales | Minimum Sales = MIN(Orders[Amount]) |
| • Delivered Orders | Delivered Orders = CALCULATE(COUNTROWS(Orders),
Orders[Delivery Status] = "Delivered") |

- Cancelled Orders Cancelled Orders = CALCULATE(COUNTROWS(Orders), Orders[Delivery Status] = "Cancelled"
- Returned Orders Returned Orders = CALCULATE(COUNTROWS(Orders), Orders[Delivery Status] = "Returned"

Charts Analysis

<u>Chart</u>	<u>Visual</u>	<u>Fields</u>	<u>Purpose</u>
Total Sales KPI	Card	Total Sales	Display total sales.
Total Orders KPI	Card	Total Orders	Display total orders.
Total Profit KPI	Card	Total Profit	Display total profit.
Total Customers KPI	Card	Total Customers	Display unique customers.
Monthly Sales Trend	Line Chart	Month, Total Sales	Analyze sales trend over time.
Sales by Category	Clustered Column Chart	Category, Total Sales	Compare sales across categories.
Profit by Region	Bar Chart	Region, Total Profit	Compare profit by region.
Top 10 Customers	Bar Chart	Customer Name, Total Sales	Identify top customers.
Sales by Payment Method	Pie Chart	Payment Method, Total Sales	Show revenue distribution by payment method.
Order Status	Donut Chart	Delivery Status, Order Count	Show Delivered, Cancelled, Returned orders.
Profit vs Sales	Scatter Chart	Sales, Profit	Analyze the relationship between sales and profit.
Sales by Sub-Category	Treemap	Sub-Category, Total Sales	Compare sales across sub-categories.
Sales by State	Filled Map	State, Total Sales	Visualize geographic sales performance.

<u>Chart</u>	<u>Visual</u>	<u>Fields</u>	<u>Purpose</u>
Sales Performance Matrix	Matrix	Category, Sub-Category, Sales, Profit	Detailed performance analysis.

Insights:

Total Orders: 1K

The business processed **1,000 orders** during the analysis period, indicating a healthy order volume and consistent customer activity.

Total Customers: 250

250 unique customers placed orders, resulting in an average of **4 orders per customer**, suggesting a good level of repeat purchases.

Total Sales: ₹383K

The business generated **₹383K in total sales**, reflecting strong revenue performance and steady customer demand throughout the analysis period.

Total Profit: ₹90.22K

The business earned **₹90.22K in profit**, indicating that sales translated into positive earnings and overall profitability.

Total Target: ₹2M

The current sales of **₹383K** are significantly below the **₹2M target**, indicating that additional efforts in marketing, promotions, and customer acquisition may be needed to achieve business goals.

Delivered Orders: 455

455 orders were successfully delivered, representing the largest share of total orders and indicating efficient order fulfillment for completed purchases.

Returned Orders: 282

282 orders were returned, which is relatively high. This suggests a need to investigate common return reasons such as product quality, incorrect deliveries, or customer expectations.

Cancelled Orders: 263

263 orders were cancelled, indicating a high cancellation rate. Analyzing cancellation reasons—such as stock availability, payment failures, or delivery delays—could help improve order completion.

Overall Business Insights

- The business generated **₹383K in sales** and **₹90.22K in profit**, demonstrating profitable operations.
- Customer engagement is strong, with **1,000 orders from 250 customers**, averaging **4 orders per customer**.
- While delivered orders form the largest group, the combined **returned (282)** and **cancelled (263)** orders account for **545 orders**, which is higher than the number of delivered orders. Reducing returns and cancellations could significantly improve revenue and customer satisfaction.
- Achieving only **₹383K** against a **₹2M target** highlights a substantial performance gap, making sales growth and operational improvements key priorities.

Business Recommendations

Increase Sales Revenue:

- Offer seasonal discounts and bundle deals on high-demand products.

Reduce Order Returns:

Analyze the reasons for the **282 returned orders**.

- Improve product descriptions and images to match customer expectations.
- Quality checks before shipping.

Minimize Order Cancellations:

- Investigate why **263 orders were cancelled**.
- provide multiple secure payment options

Improve Target Achievement:

- Current sales (**₹383K**) are well below the target (**₹2M**).
- Set monthly and quarterly sales targets for each region and category.
- Monitor sales performance regularly.

Increase Customer Retention:

- Reward repeat customers.
- Send personalized offers based on customer purchase history.
- Collect customer feedback to improve satisfaction and encourage repeat purchases.

Focus on High-Performing Products:

- Promote best-selling and high-profit products through advertisements and featured placements.

Regional Performance:

- Identify regions with low sales and run location-specific promotions.
- Allocate more inventory and marketing resources to high-performing regions.

Improve Delivery Performance:

- Increase the percentage of successfully delivered orders by partnering with reliable logistics providers.
- Provide customers with real-time order tracking and estimated delivery times.

Optimize Profitability:

- Focus on products with higher profit margins.
- Review pricing strategies.

Expected Business Outcomes

- Increase sales revenue toward the ₹2M target.
- Reduce return and cancellation rates.
- Improve customer satisfaction and loyalty.
- Increase delivered orders and operational efficiency.
- Enhance overall profitability through better product and customer strategies.

Conclusion

This **E-Commerce Sales Analysis** project successfully transformed raw and inconsistent data into meaningful business insights using **Excel, Power Query, and Power BI**. Data quality issues such as missing values, duplicate records, inconsistent text formatting, and invalid entries were identified and resolved to ensure accurate analysis.

An interactive Power BI dashboard was developed to monitor key performance indicators, including **Total Sales (₹383K), Total Profit (₹90.22K), Total Orders (1,000), Total Customers (250), Sales Target (₹2M), Delivered Orders (455), Returned Orders (282), and Cancelled**

Orders (263). The dashboard also provided insights into sales trends, customer behavior, product performance, regional performance, and order status.

The analysis revealed opportunities to improve business performance by increasing sales, reducing order cancellations and returns, strengthening customer retention, and focusing on high-performing products and regions. These insights enable stakeholders to make informed, data-driven decisions and support future business growth.

Overall, this project demonstrates the complete data analytics workflow—from data exploration and cleaning to modeling, visualization, and business recommendation—showcasing practical skills in **Excel, Power Query, DAX, and Power BI** for solving real-world business problems.

